

2. Universal Approximation Theorem (15 marks)

(a) Universal Approximation Theorem (Formal Statement)

The Universal Approximation Theorem (UAT) states that a feedforward neural network with a single hidden layer and a non-polynomial continuous activation function can approximate any continuous function on a compact domain to arbitrary accuracy.

Formal Statement

Let:

- $K \subset \mathbb{R}^d$ be a compact set (i.e., closed and bounded),
- $f \in C(K)$, meaning $f : K \rightarrow \mathbb{R}$ is a continuous function on K ,
- $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous, non-polynomial activation function (e.g., sigmoid or tanh).

Then, for every $\varepsilon > 0$, there exists:

- an integer $N \in \mathbb{N}$,
- weights $w_i \in \mathbb{R}^d$,
- biases $b_i \in \mathbb{R}$,
- output weights $\alpha_i \in \mathbb{R}$,

such that the function

$$g(x) = \sum_{i=1}^N \alpha_i \sigma(w_i^\top x + b_i)$$

satisfies

$$\|f - g\|_\infty < \varepsilon,$$

where

$$\|f - g\|_\infty = \sup_{x \in K} |f(x) - g(x)|.$$

Required Specifications

1. Class of Target Functions The theorem applies to all continuous functions

$$f \in C(K)$$

defined on a compact subset $K \subset \mathbb{R}^d$.

2. Assumptions on the Domain The domain K must be compact, meaning:

- Closed,
- Bounded.

A common example is

$$K = [0, 1]^d.$$

3. Notion of Approximation and Norm Used The approximation is in the uniform (supremum) norm, defined as:

$$\|f - g\|_\infty = \sup_{x \in K} |f(x) - g(x)|.$$

This means the maximum absolute difference between f and the neural network approximation g over the entire domain is less than ε .

(b) Main Idea Behind the Proof of the Universal Approximation Theorem

A full formal proof of the Universal Approximation Theorem (UAT) is technical, but the central idea can be understood through two key perspectives:

- Constructive approximation using “basis-like” functions,
- Connection to classical approximation theory (Stone–Weierstrass theorem).

1. Constructive Approximation Intuition

A single-hidden-layer neural network has the form

$$g(x) = \sum_{i=1}^N \alpha_i \sigma(w_i^\top x + b_i).$$

Each hidden neuron computes a function of the form

$$\sigma(w^\top x + b),$$

which can be interpreted as a soft thresholded hyperplane.

Key Idea: By adjusting w and b , each neuron can:

- Create a “bump”-like response in specific regions of the input space,
- Approximate localized features of a function.

By taking linear combinations of many such neurons (via output weights α_i), we can:

- Combine many localized responses,
- Approximate increasingly complex shapes.

This is similar to representing a function using:

- Fourier series,
- Polynomial bases,
- Radial basis functions.

In other words, sigmoid neurons act like basis functions, and increasing the number of hidden units increases the expressive power of the approximation.

As $N \rightarrow \infty$, these combinations can approximate any continuous function on a compact domain arbitrarily well.

2. Connection to the Stone–Weierstrass Theorem

A deeper theoretical argument connects neural networks to classical approximation theory.

Classical Result: The Stone–Weierstrass Theorem states that certain families of functions are dense in $C(K)$, the space of continuous functions on a compact set K . Density means that any continuous function can be uniformly approximated by functions from that family.

Key Theoretical Insight: The set of functions

$$\mathcal{H} = \left\{ \sum_{i=1}^N \alpha_i \sigma(w_i^\top x + b_i) \right\}$$

can be shown to be:

- An algebra of functions,
- Capable of separating points,
- Closed under linear combinations.

If the activation function σ is continuous and non-polynomial, then this family satisfies the conditions required for density in $C(K)$.

Therefore,

$$\overline{\mathcal{H}} = C(K)$$

in the uniform norm.

This establishes that single-hidden-layer networks are universal approximators.

3. Why “Non-Polynomial” is Important

If the activation function were polynomial, then the network would only generate polynomial functions of bounded degree, which are not dense in $C(K)$ unless the degree grows without bound.

A non-polynomial activation ensures the function class is rich enough to generate arbitrarily complex nonlinear behavior.

(c) Two Important Limitations of the Universal Approximation Theorem

1. The Theorem Is an Existence Result, Not a Learnability Result

Limitation The UAT states that a neural network can represent any continuous function on a compact set, but it does not guarantee that:

- We can find the required weights using training algorithms (e.g., gradient descent),
- Training will be computationally feasible,
- Optimization will converge to a good solution.

The theorem only proves the existence of suitable parameters — not that they are accessible in practice.

Why This Matters in Practice In real machine learning:

- Training involves solving a non-convex optimization problem.
- Loss surfaces may contain local minima, saddle points, or flat regions.
- Gradient-based methods may fail to reach a good approximation even though one theoretically exists.

Thus, even though a network is theoretically capable of approximating a function, it may:

- Train very slowly,
- Get stuck in poor solutions,
- Require careful initialization and tuning.

Practical implication: Architecture design, initialization strategies, optimization methods, and regularization techniques are crucial — expressive power alone is not sufficient.

2. No Guarantee on Network Size or Efficiency

Limitation The UAT does not specify:

- How many hidden units N are required,
- How large the weights must be,
- Whether the required network size is practical.

In fact, for some functions, a single-hidden-layer network may require an exponentially large number of neurons in the input dimension d .

Why This Matters in Practice In high-dimensional problems:

- A shallow network may need an extremely large width to approximate complex functions.

This leads to:

- High computational cost,
- Large memory requirements,
- Increased risk of overfitting,
- Slow training.

This limitation partly explains why deep networks are preferred in practice:

- Depth can represent certain functions more efficiently than width.
- Some functions that require exponentially many neurons in shallow networks can be represented with polynomial size using deeper architectures.

Practical implication: The theorem does not justify using shallow networks in real-world tasks. Architectural choices (depth vs. width) strongly affect efficiency and generalization.

(d) Universal Approximation with ReLU and Increasing Depth

When using ReLU activations and allowing the depth of the network to increase (while keeping the width relatively small), the nature of the universal approximation property changes in an important and practically significant way.

1. ReLU Networks Are Also Universal Approximators

Although ReLU is piecewise linear, it is non-polynomial, and therefore:

- A sufficiently wide single-hidden-layer ReLU network can approximate any continuous function on a compact set.
- Hence, the universal approximation property still holds.

However, the more interesting result concerns depth.

2. Depth Can Replace Width

For ReLU networks, it has been shown that:

- A network with width at least $d + 1$ (where d is the input dimension),
- And arbitrary depth,

is a universal approximator.

This means that instead of making the network extremely wide (as in the classical single-layer theorem), we can:

- Keep the width relatively small,
- Increase the depth,
- Still approximate any continuous function on a compact domain.

3. Why This Is Important

(a) Improved Efficiency Certain functions that require exponentially many neurons in a shallow network can be represented with:

- Polynomial size,
- Much fewer parameters,

when depth is increased.

Thus, depth provides representational efficiency.

(b) Compositional Structure Deep ReLU networks naturally build:

- Piecewise linear functions,
- With exponentially many linear regions as depth increases.

Each additional layer increases the number of linear regions the network can represent, allowing highly complex functions to be approximated using relatively small width.

4. Practical Interpretation

Compared to the classical UAT for shallow sigmoid networks:

- Shallow networks achieve universality through large width.
- Deep ReLU networks achieve universality through depth.

This explains why modern deep learning prefers:

- Moderately wide,
- Deep ReLU networks,

rather than extremely wide single-hidden-layer networks.